



SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING



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QUESTIONS

20 / 20 completed

Shape-2

SHAPE-2

Write a Java program to implement a basic shape hierarchy using polymorphism. Create a parent class called Shape with attributes such as area and perimeter. Then, create child classes such as Circle and Rectangle that inherit from the Shape class and have their own unique methods for calculating area and perimeter.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     public static void main(String[] args) {
3         // Polymorphism: parent reference points to child objects
4         Shape shape1 = new Circle(5.0);
5         Shape shape2 = new Rectangle(4.0, 6.0);
6
7         shape1.display();
8         System.out.println();
9         shape2.display();
10    }
11 }
12
13 class Shape {
14     double area;
15     double perimeter;
16
17     void display() {
18         System.out.println("Area: " + area);
19         System.out.println("Perimeter: " + perimeter);
20     }
21 }
```

OUTPUT

Your OUTPUT will appear here!



INPUT

Your INPUT goes here!

